



Education

University of Michigan, Ann Arbor

B.S. IN COMPUTER SCIENCE (ROBOTICS & VISION TRACK) + PHILOSOPHY

GPA: 4.0/4.0 (A+ for all Major courses)

• Grad Level Courses: Autonomous Robotics, Reinforcement Learning, Mobile Robotics, Action and Perception

Advisor: Prof. Maani Ghaffari, Prof. Stella Yu @ UMich | Prof. Mengdi Wang @ Princeton

Ann Arbor, MI

Jan 2025 - Dec 2026

University of Waterloo (transferred out)

BACHELOR OF MATHEMATICS

GPA: 3.9

Waterloo, ON

Aug 2021 - May 2024

Research Interest

My research interests lie in [Embodied AI](#), [Multimodal Robot Learning](#), and [Neuro-Symbolic AI](#), drawing on [Cognition](#): how the human mind builds a stable, predictive model of the physical world and reuses what it knows to act reliably. I develop scalable [World Models](#) that capture physical dynamics through robust latent state estimation, and study how capabilities latent in foundation models can be elicited and steered into reliable behavior. I aim to bridge [multimodal perception and embodied spatial understanding](#) to create actionable representations for reliable robotic decision-making and physical interaction.

Publications

WORKS IN PROGRESS

[1] Learning Whole-Body Human–Humanoid Collaboration from Egocentric Vision via Multi-Agent Distillation

Co-first author

[In Preparation for CVPR 2027](#)

[2] Eliciting and Steering Latent Capabilities in VLA Models with Neuron-Targeted Training

First author

[In Preparation for CVPR 2027](#)

CONFERENCE PUBLICATION HIGHLIGHTS (SEE ALL PAPERS IN GOOGLE SCHOLAR)

[1] LaWM: Least Action World Models for Long-Horizon Physical Consistency from Visual Observations.

Q. Xiao[†], M. Ghaffari

[Submitting to CVPR 2027](#), [\[Paper\]](#) · [\[Project Page\]](#) · [\[GitHub\]](#)

[2] MACD: Model-Aware Contrastive Decoding via Counterfactual Data for Video-LLMs.

Q. Xiao[†], K. Zhou

[ICML 2026](#), [\[Paper\]](#)

[3] GenEnv: Difficulty-Aligned Co-Evolution Between LLM Agents and Environment Simulators.

J. Guo*, L. Yang*, Q. Xiao*, ..., M. Wang[†] (Co-first author)

[NeurIPS 2026 \(and ICLR 2026 Workshop\)](#), [\[Paper\]](#) · [\[GitHub\]](#)

[4] LongNav-R1: Horizon-Adaptive Multi-Turn RL for Long-Horizon VLA Navigation.

Y. Hu, A. Xi, Q. Xiao, ..., R. Vasudevan[†], M. Ghaffari[†]

[RSS 2026](#), [\[Paper\]](#)

[5] MoPe: Motion Permanence for Robust Monocular Gaussian Mapping in Dynamic Environments.

Q. Xiao

[RSS Workshop 2026](#), [\[Paper\]](#)

[6] CCAD: Anomaly Detection Conditioned on a Compressed Global Feature Space.

X. Jin*, L. Diao*, Q. Xiao, Y. Hu, Y. Liu, Z. Zhang, H. Gu

[Preprint](#), [\[Paper\]](#) · [\[GitHub\]](#)

Research Experience

Lead-Authored Research

FIRST & CORRESPONDING AUTHOR

MACD: Model-Aware Contrastive Decoding via Counterfactual Data for Video-LLMs

ICML 2026

- Proposed MACD, a training-free inference strategy that reduces **Video-LLM hallucination** by integrating **model-aware counterfactual inputs** into contrastive decoding for evidence-grounded token selection.
- Built an object-/frame-level counterfactual pipeline: detect and track objects, initialize **soft spatio-temporal masks**, then optimize mask strengths via **gradient ascent on the VLM loss** to mine evidence-critical regions.

Eliciting and Steering Latent Capabilities in VLA Models with Neuron-Targeted Training

In Preparation for CVPR 2027

- Overcoming the catastrophic forgetting associated with full-model fine-tuning by developing a **neuron-targeted training** framework for VLA models.
- Identifying capability-specific neurons in foundation models to contrastive success/failure datasets.
- Correcting specific failures via **localized updates**, keeping the rest of the pre-trained model behaviors strictly intact.

University of Michigan (Stella Yu's Group)

Ann Arbor, MI

RESEARCH INTERN

Sep 2026 - Present

Learning Whole-Body Human-Humanoid Collaboration from Egocentric Vision via Multi-Agent Distillation

In Preparation for CVPR 2027

- Eliminating motion capture by directing human-robot collaboration entirely through **onboard egocentric vision**.
- Translating multi-agent RL policies trained in state space into deployable **vision-based models** via policy distillation.
- Scaling robot control from restricted upper-body commands to full **whole-body coordination** to support more complex physical interactions.

University of Michigan (Maani Ghaffari's Group)

Ann Arbor, MI

RESEARCH INTERN

Oct 2025 - Present

LaWM: Least Action World Models for Long-Horizon Physical Consistency from Visual Observations

Submitting to CVPR 2027

- Proposed **LaWM**, a latent world-modeling framework that operationalizes the **Principle of Least Action** in learned visual latent space, making physical structure part of the rollout mechanism rather than a post-hoc regularizer.
- Designed a **latent variational integrator** that encodes observations into generalized latent coordinates, learns a **latent discrete Lagrangian**, constructs a discrete action functional, and advances prediction through the **discrete Euler-Lagrange** condition.
- Built an end-to-end visual prediction system with learned encoder, decoder, latent Lagrangian, context encoder, and differentiable finite-step DEL solver, enabling long-horizon rollout directly from visual observations.

LongNav-R1: Horizon-Adaptive Multi-Turn RL for Long-Horizon VLA Navigation

RSS 2026

- Proposed **LongNav-R1**, an end-to-end **Multi-Turn RL** framework that reformulates long-horizon 3D embodied VLA navigation as a continuous **Conversation** between policy and environment, with an **implicit latent state** in context.
- Developed Horizon-Adaptive mechanism (HAPO) with **critic-free kernel regression** and **temporal-aware kernels**, for long-horizon credit assignment over Causal Effects and Sequential Future Outcomes beyond single-turn BC.
- Optimized **VLA** models via scalable multi-turn rollouts with KV-cache reuse and online token pruning, supporting efficient long-context decision-making in stochastic 3D environments.

Princeton University (Mengdi Wang's Group)

Princeton, NJ

RESEARCH INTERN

May 2025 - Oct 2025

GenEnv: Difficulty-Aligned Co-Evolution Between LLM Agents and Environment Simulators

NeurIPS 2026 (and ICLR 2026 Workshop)

- Developed **GenEnv**, a co-evolutionary framework where an **LLM-based Simulator** acts as a curriculum simulator to generate **difficulty-aligned curricula**, shifting training from static supervision to a **Data-Evolving Paradigm**.
- Designed the α -Curriculum Reward to align task difficulty with the agent's Zone of Proximal Development, forming a **co-evolution loop** where the environment evolves to consistently challenge the agent.
- Implemented **Reward-Weighted Regression** for closed-loop simulator updates, selectively prioritizing high-value training trajectories.